

26**A**

Magnetic flux linkage in the coil at position X
 $= NBA$

Magnetic flux linkage in the coil at position Y
 $= 0$

By Faraday's law, average e.m.f. induced in the coil during the rotation

$$\langle E \rangle = \frac{\Delta \Phi}{\Delta t}$$

$$= \frac{3000 \times 1.8 \times \frac{\pi}{4} \times 0.020^2}{0.060}$$

$$= 28.3$$

$$\approx 28 \text{ V}$$